

Bana's Green Notebook

A student's knowledge base on carbon footprints in Nepal — source for the Bana AI chatbot.

Carbon Footprint Basics

What is a carbon footprint?

A carbon footprint is the total greenhouse gases your activities release, measured as kilograms of carbon dioxide (CO₂). Travel, electricity, cooking fuel, food and waste all add to it. For a Nepali school student the biggest parts are usually transport, cooking fuel and food. You lower it by choosing cleaner options each day.

Keywords: carbon footprint, CO₂, meaning, basics, ████████, ██████████

Greenhouse gases and units

Carbon dioxide (CO₂) comes from burning petrol, diesel, firewood and LPG. Methane (CH₄) comes from livestock, rice paddies and rotting waste, trapping far more heat per kilogram. Footprints are measured in kilograms or tonnes of CO₂-equivalent. In Nepal about 54 percent of emissions are methane, mostly from agriculture.

Keywords: greenhouse gas, methane, CH₄, CO₂, units, ████████, ██████

Nepal emits little but suffers a lot

Nepal emits little: in 2021 about 1.9 tonnes of CO₂ per person, while the global average was about 6.9. Yet Nepal suffers strongly through melting glaciers, floods and erratic monsoons. So Nepali students reduce their footprint not from guilt, but to protect their mountains and rivers and show leadership.

Keywords: Nepal, per capita, vulnerable, climate justice, ████████████████, ████████

Nepal Climate Context

How climate change hits Nepal

Nepal is warming faster than the global average, especially in the high mountains. Glaciers are shrinking and dangerous glacial lakes are growing; Nepal has recorded about 26 glacial-lake outburst floods. Black carbon (soot) from firewood and diesel settles on snow and speeds melting. Monsoon rain is more erratic, causing floods and landslides.

Keywords: glaciers, GLOF, Himalaya, monsoon, floods, ████████, ██████, ██████

Air pollution in Nepal's cities

Kathmandu Valley traps polluted air because it is a bowl ringed by hills. Main sources are vehicle exhaust, brick kilns, road and construction dust, and open burning of waste and stubble. Transport alone causes around half of Nepal's energy-related CO₂. Cleaner transport, no open burning and dust control help the valley breathe.

Keywords: Kathmandu, air pollution, brick kiln, vehicles, dust, ██████████, ██████████, ██████████

Nepal's climate goals

Nepal has promised net-zero carbon emissions by 2045 and to keep at least 45 percent of the country under forest. Its plan aims for many more electric vehicles and for 25 percent of households to use electric stoves as their main cooking method by 2030, plus many improved cookstoves and biogas plants.

Keywords: net zero 2045, NDC, forest cover, clean cooking, ■■■ ■■■■, ■■, ■■■■■■

Why Nepal's electricity is clean

Almost all of Nepal's grid electricity comes from hydropower, so it is nearly 100 percent renewable and low-carbon. About 98 percent of people have electricity, and in the wet season Nepal has surplus power and exports some to Bangladesh. So switching from fuel (firewood, LPG, petrol) to grid electricity usually cuts your footprint.

Keywords: hydropower, electricity, renewable, grid, ■■■■■■■■, ■■■■■■, ■■■■■■■■

Transport

Transport is Nepal's biggest energy emitter

Burning petrol and diesel causes about half of Nepal's energy-related CO₂, and petroleum is one of the largest, costliest imports. Each litre of petrol burned makes about 2.3 kg of CO₂. So how you travel is often the single biggest lever a student has. Feet, a cycle, shared rides or public transport beat a private motorbike or car.

Keywords: transport, petrol, diesel, petroleum, ■■■■■■■■, ■■■■■■■■, ■■■■■■

Cleanest to dirtiest ways to travel

Cleanest to most polluting for a student: walking and cycling are zero-emission and healthy; a shared public bus or microbus (like Sajha Yatayat) spreads its fuel over many riders, so per-person share is small; a motorbike is worse per person; a private car with one student is the most polluting. Sharing any vehicle lowers the footprint per person.

Keywords: walk, cycle, bus, microbus, Sajha, motorbike, car, ■■■■■■■■, ■■■■■■, ■■

Electric vehicles in Nepal

Nepal is a world leader in EV adoption: in 2024/25 EVs were about 73 percent of four-wheeler passenger imports, helped by low EV taxes. Because the grid is hydropower, an EV runs on clean energy with a low footprint. There are around 750 charging stations, Kathmandu uses electric garbage trucks, and Bagmati requires new taxis to be electric.

Keywords: EV, electric vehicle, hydropower, charging, ■■■■■■■■ ■■■■■■, ■■■■■■■■

Travel tips for students

Walk or cycle for trips under 2-3 km: free, healthy and zero-carbon. For longer trips take a public bus or microbus instead of a private vehicle, and share rides with classmates nearby. Combine errands into one trip. Ask family about an electric scooter or electric public transport. Keeping tyres pumped and engines serviced saves fuel too.

Keywords: school commute, walk, cycle, share, ■■■■■■, ■■■■■■■■, ■■■■

Home Energy

Using clean grid electricity wisely

Because Nepal's electricity is hydropower, using grid power for light, fans, study and cooking is already low-carbon. Switching a home from firewood or LPG to electric usually cuts the footprint. Still, save energy: power you don't waste can run someone else's induction stove or an electric bus, and it lowers your bill and eases the dry-season grid.

Keywords: electricity, hydropower, energy saving, home, ■■■■■■■■, ■■■■■■ ■■■■

Easy ways to save energy at home

Switch to LED bulbs (up to 80 percent less power). Turn off lights, fans and the TV when you leave a room, and unplug chargers that sip standby power. Study near a window for daylight. Use a fan instead of air-conditioning. Dry clothes in the sun. These habits cost nothing and free up clean power for everyone.

Keywords: LED, lights, fans, unplug, natural light, ■■■■■■, ■■■■■■, ■■■■■■

Solar power for homes and schools

Nepal gets sunshine on about 300 days a year, so rooftop solar panels and solar water heaters work well, especially where the grid is weak. A small solar panel and a rechargeable solar lantern give clean study light without kerosene smoke. Encourage your school to put solar panels on its roof; it makes clean power on-site and is a great eco-club project.

Keywords: solar, rooftop, solar lantern, sun, ■■■■■■, ■■, ■■■■■■

Cooking Fuel

Choosing a cleaner cooking fuel

Cooking fuel is a big part of a household's footprint. Cleanest to dirtiest for climate and lungs: an electric induction stove (on hydropower) is best; biogas from animal or kitchen waste is very good; LPG gas is cleaner than wood; open firewood is the most harmful. Moving even one or two meals a day to induction or biogas makes a real difference.

Keywords: cooking, firewood, LPG, biogas, induction, ■■■■■■, ■■■■■■, ■■■■■■■■

Why firewood is harmful

Open firewood cuts forests, fills kitchens with smoke that harms eyes and lungs (especially mothers and children), and releases black carbon that lands on Himalayan snow and speeds melting. An improved cookstove burns wood far more efficiently and vents smoke outside, while induction or biogas removes smoke entirely. Reducing firewood protects health and the mountains.

Keywords: firewood, deforestation, indoor smoke, black carbon, ■■■■■■, ■■■■■■, ■■ ■■■■■■

Clean cooking options in Nepal

Nepal aims for a quarter of households to cook mainly with electricity by 2030, plus many biogas plants and improved cookstoves. Because the grid is hydropower, an induction stove is clean and cheap to run. Biogas turns cow dung and kitchen scraps into cooking gas and leaves good fertiliser, perfect for Terai and hill farms. An improved cookstove is a low-cost first step.

Keywords: induction, biogas, improved cookstove, NDC, ■■■■■■■■, ■■■■■■■■, ■■■■■■■■ ■■■■

Food

Food and your footprint

Food can be a quarter of a person's footprint. Locally grown, seasonal, plant-based meals are low-carbon, while meat, dairy and packaged or imported foods are higher due to methane, processing and long transport. For a Nepali student, eating fresh local food and cutting packaged snacks is an easy, tasty win.

Keywords: food, footprint, local, seasonal, packaged, ■■■■■, ■■■■■■■■, ■■■■■■

Dal bhat and local food

A simple dal-bhat-tarkari from local, seasonal vegetables and lentils is low-carbon and healthy. Packaged noodles, chips, soft drinks and imported snacks have a much higher footprint from processing, packaging and long transport. Choosing local seasonal produce and home food over packaged snacks cuts both carbon and plastic.

Keywords: dal bhat, local, seasonal, packaged snacks, ■■■■■■, ■■■■■■■■, ■■■■

Meat, dairy and methane

Buffalo, cows and goats burp methane, a powerful greenhouse gas, and need much land and feed. You need not stop meat, but having it fewer times a week and in smaller portions lowers your footprint. Plant proteins common in Nepal — dal (lentils), beans, soybean and gundruk with rice — are filling, cheap and low-carbon.

Keywords: meat, buff, chicken, goat, dairy, methane, ■■■■, ■■■■■■■■, ■■■■■■

Cut food waste, cut methane

When food rots in a dump it releases methane. Take only what you will eat, store leftovers and share extra food. Peels and scraps should be composted, not thrown in mixed waste. Since a large share of Nepal's emissions is methane, reducing food waste and composting scraps is one of the most effective household actions.

Keywords: food waste, compost, methane, portions, ■■■■ ■■■, ■■■■■■■■■■, ■■■■■■

Grow a kitchen garden

A home or school kitchen garden (karesabari) means food travels almost no distance, needs no packaging, and scraps compost right there. Herbs, greens, tomatoes and beans grow well in pots or small plots across Nepal. A school vegetable garden is a fun eco-club project that teaches where low-carbon food comes from.

Keywords: kitchen garden, karesabari, vegetables, school garden, ■■■■■■■■■■, ■■■■■■■■

Waste

Managing waste in Nepal

Separate waste into wet (organic) and dry (recyclable) at home and school. Nepal's Solid Waste Management Act (2011) supports proper management, but enforcement is weak, so personal habits matter. Some Kathmandu municipalities use electric garbage trucks. Segregating lets organic matter become compost and paper, glass and metal be recycled instead of dumped or burned.

Keywords: waste, segregation, organic, recyclable, ■■■■■■, ■■■■■■■■■■, ■■■■■■

Composting organic waste

Composting turns kitchen scraps, peels and garden trimmings into rich fertiliser instead of sending them to a dump where they release methane. A simple compost pit or bin in the yard, or a school compost corner, does the job. The finished compost feeds a kitchen garden, closing the loop — one of the cheapest, highest-impact climate actions.

Keywords: compost, organic, methane, fertiliser, ■■■■■■■■■■, ■■■, ■■■■■■

Reducing plastic

Carry a cloth or jute bag and refuse free plastic bags, which Kathmandu Valley has tried to ban. Use a refillable steel water bottle instead of plastic bottles. Choose less packaging and reuse containers. Plastic is often burned or dumped in Nepal because recycling is limited, so avoiding single-use plastic is the best

step.

Keywords: plastic, single-use, cloth bag, refill, ■■■■■■■■■■, ■■■■■■■■■■

Never burn your trash

Burning rubbish or crop stubble releases toxic smoke and black carbon that pollutes the air and darkens Himalayan snow, speeding melting. It is one of the worst things for health and climate. Instead, compost organic waste, recycle dry waste, and hand the rest to municipal collection. Encourage neighbours and farmers not to burn.

Keywords: open burning, black carbon, toxic smoke, stubble, ■■■■, ■■■■ ■■■■■■■■, ■■■■

Water and Forests

Saving water

Treating and pumping water uses energy, and clean water is precious in much of Nepal. Turn off taps while brushing, fix drips, and reuse vegetable-washing water for plants. A simple rooftop rainwater-harvesting tank stores monsoon rain for dry months and reduces pumping. A bucket and mug save a surprising amount.

Keywords: water, conservation, rainwater harvesting, leaks, ■■■■, ■■■■■■■■■■

Trees absorb carbon

A growing tree pulls CO₂ from the air; a healthy tree can absorb roughly 20 kg of CO₂ a year. Native Nepali species such as sal, utis (alder), oak (kharsu) and rhododendron (lali gurans) suit local conditions and support birds and bees. Planting and caring for trees until well established is a powerful, lasting climate action.

Keywords: trees, carbon sink, native species, sal, utis, ■■■■, ■■■■■■■■, ■■■■■■■■■■

Nepal's famous community forests

Nepal's community-forestry programme is admired worldwide: local community forest user groups protect and manage nearby forests, helping raise forest cover back to around 45 percent. Joining or supporting a community forest, helping with plantation days, and preventing forest fires all strengthen this natural carbon sink and show how ordinary people can heal the land.

Keywords: community forestry, forest user group, restoration, ■■■■■■■■■■ ■■, ■■■■■■■■■■ ■■■■

Taking Action

Set a SMART eco goal

Make your climate goal SMART: Specific, Measurable, Achievable, Relevant, Time-bound. Instead of 'I will save the planet', try 'This week I will walk to school three days.' Pick one small action, do it for a week, then add another. Small steady habits beat big promises. Track your wins to stay motivated.

Keywords: SMART goal, plan, weekly, habit, ■■■■■■■■■■ ■■■■■■■■, ■■■■■■■■, ■■■■

Climate action at school

Schools multiply your impact. Start or join an eco-club, organise a tree-planting and tree-care day, set up waste-segregation and compost corners, switch the school to LED lights, and run awareness skits or poster days. Ask about solar panels for the roof. A class-versus-class green challenge makes saving carbon fun.

Keywords: eco club, school, tree planting, segregation, ■■■■ ■■■■, ■■■■■■■■, ■■■■■■■■■■

Lead at home and with friends

You influence more people than you think. Explain to family why you switch off lights, carry a cloth bag or want induction cooking, and lead by example rather than lecturing. Share one tip with a friend each week. When young people act kindly and consistently, parents, shopkeepers and neighbours often follow.

Keywords: family, advocacy, role model, friends, ■■■■■■, ■■■■, ■■■■■■

Measure, then improve

You cannot improve what you do not measure. Use a carbon-footprint calculator to find your biggest source, set one goal to shrink it, then re-check after a few weeks. Celebrate small drops and share them with your class. Tracking turns a vague wish into visible, motivating results.

Keywords: measure, track, calculator, progress, ■■■■, ■■■■■■■■■■, ■■■■■■

How much do actions save?

Rough savings: each litre of petrol not burned saves about 2.3 kg CO₂; walking or cycling a 2 km school trip instead of a motorbike saves roughly 0.5 kg each way; moving a meal from firewood or LPG to induction cuts smoke and carbon daily; composting and avoiding open burning cut methane and black carbon; one well-cared tree absorbs about 20 kg CO₂ a year.

Keywords: savings, numbers, kg CO₂, reference, ■■■■, ■■■■

Common Questions

Q: How can I reduce my carbon footprint?

Start with your biggest source — for most students that is transport, so walk or cycle short trips and take a bus or shared ride for longer ones. Next, push for cleaner cooking (induction on hydropower, or biogas) and eat local seasonal food with less packaging and less meat. Segregate and compost waste, never burn trash, save power and water, and plant a native tree. Pick one habit, do it a week, then add another.

Keywords: reduce footprint, how, student, plan, ■■■■■■, ■■■■, ■■■■■■

Q: Is electricity clean in Nepal?

Yes. Nearly all of Nepal's grid electricity comes from hydropower, so it is low-carbon and renewable. That is why switching from firewood, LPG or petrol to electric options (induction stoves, electric vehicles) usually lowers your footprint. The main thing is to avoid wasting electricity, especially in the dry season when rivers are lower.

Keywords: electricity clean, hydropower, ■■■■■■ ■■■■, ■■■■■■■■■■

Q: Why is cooking on firewood bad?

Open firewood harms forests, fills the kitchen with harmful smoke, and produces black carbon that settles on Himalayan glaciers and speeds melting. Switching to an improved cookstove, biogas or an electric induction stove protects your family's health and the mountains.

Keywords: firewood bad, smoke, deforestation, ■■■■■■ ■■■■■■■■■■, ■■■■■■

Q: What trees should I plant in Nepal?

Choose native species suited to your area, such as sal, utis (alder), oak (kharsu) or rhododendron (lali gurans). They survive better, support local birds and insects, and store carbon. Planting is only half the job; water and protect the sapling until it is well established.

Keywords: what trees, native species, plant, ■■■■ ■■■■, ■■■■■■■■■■

Q: What is the best way to get to school?

Walking or cycling is best for short trips: zero carbon, free and healthy. For longer distances, a public bus or microbus shared with many riders is far cleaner per person than a private motorbike or car. Sharing a ride with classmates is the next best thing.

Keywords: best transport school, walk, cycle, bus,                                                    